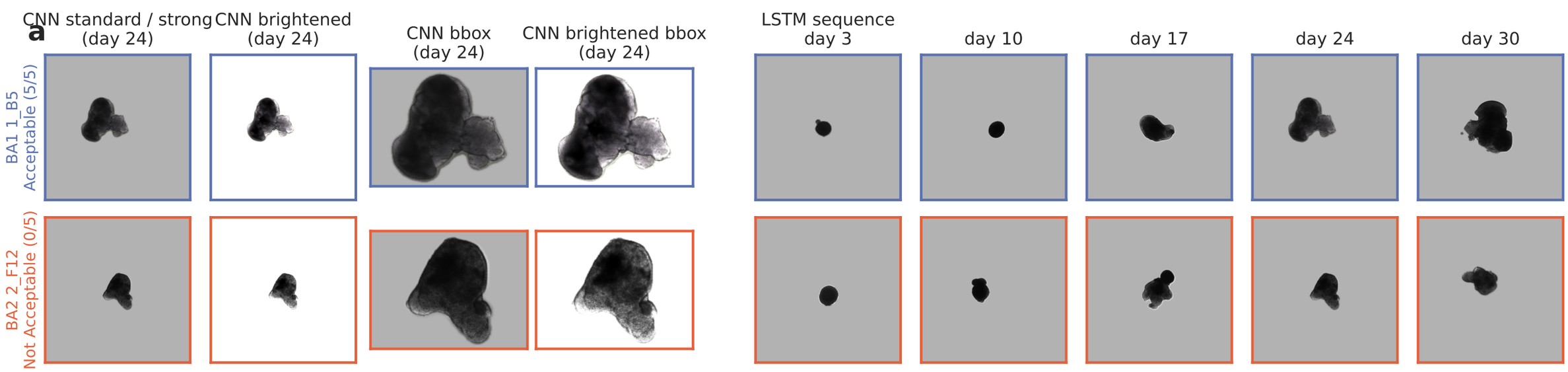
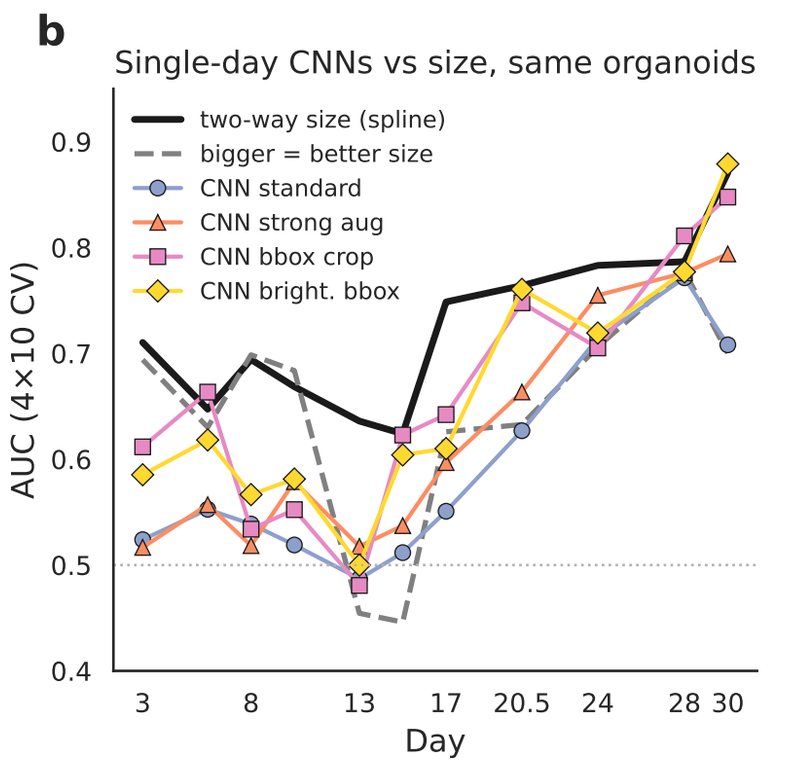
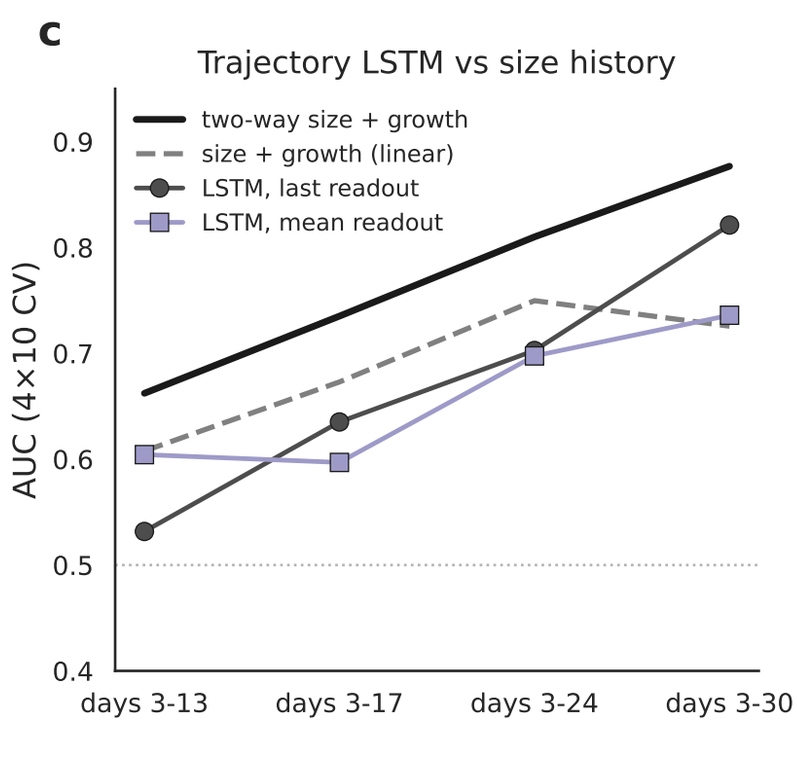
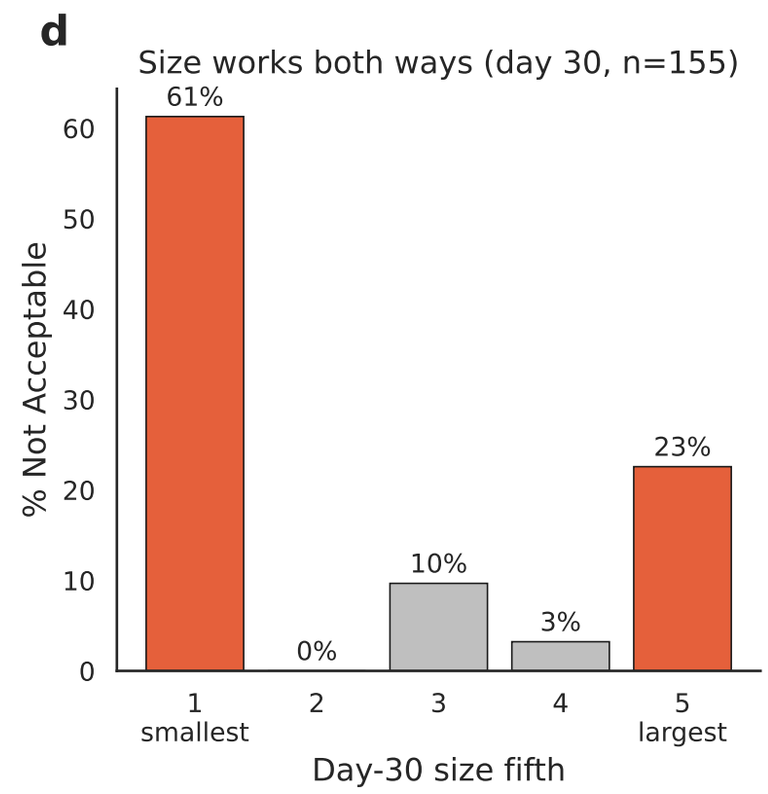
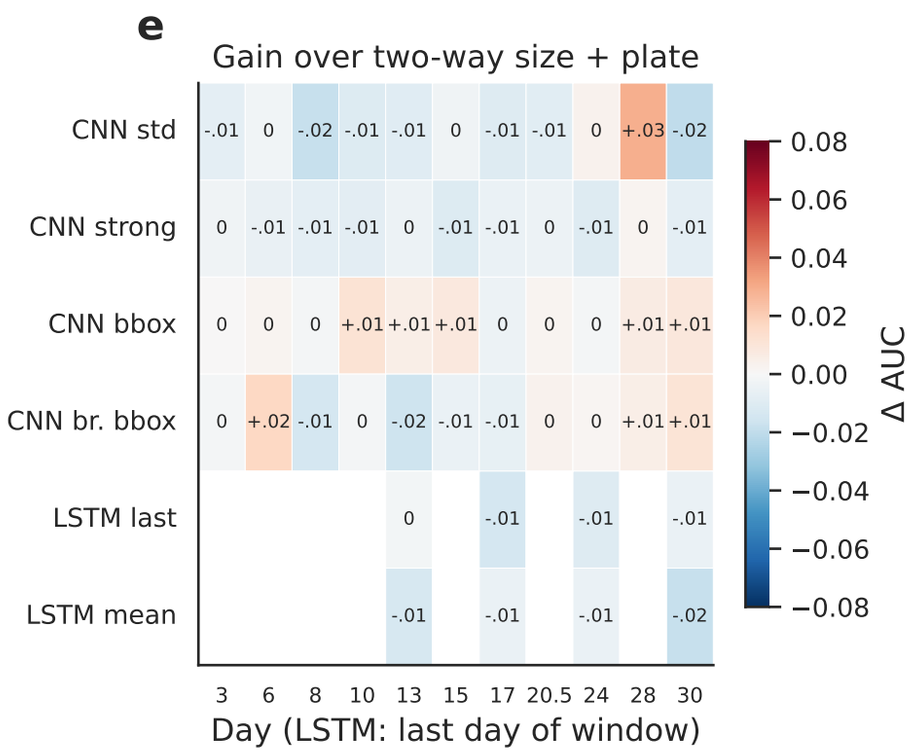
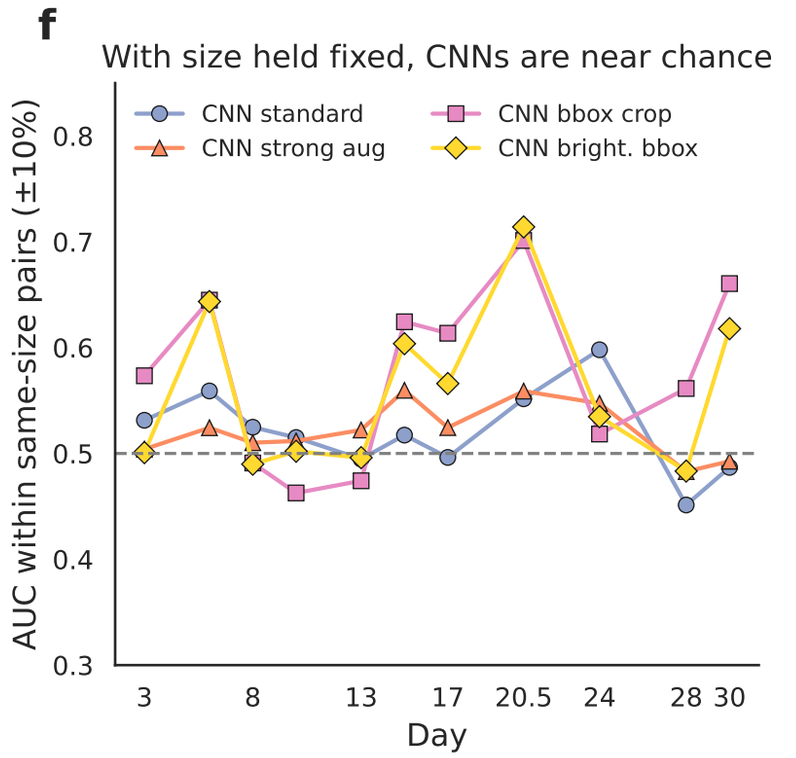

a
CNN standard / strong (day 24)
CNN brightened (day 24)
CNN bbox (day 24)
CNN brightened bbox (day 24)
LSTM sequence day 3
day 10
day 17
day 24
day 30
BA1 1_B5 Acceptable (5/5)
BA2 2_F12 Not Acceptable (0/5)

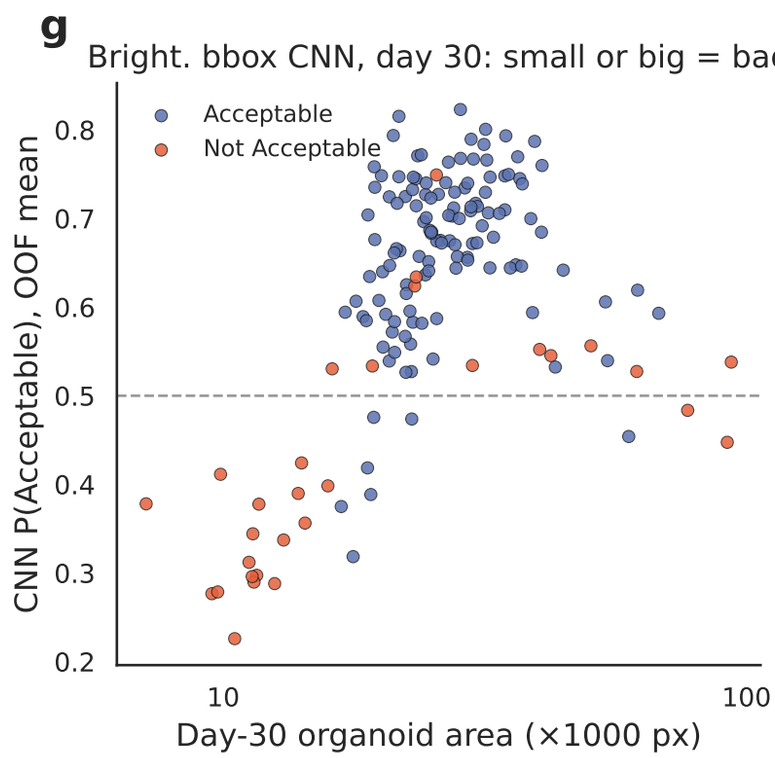

b
Single-day CNNs vs size, same organoids
two-way size (spline)
bigger = better size
CNN standard
CNN strong aug
CNN bbox crop
CNN bright. bbox
AUC (4×10 CV)
Day


c
Trajectory LSTM vs size history
two-way size + growth
size + growth (linear)
LSTM, last readout
LSTM, mean readout
AUC (4×10 CV)
days 3-13
days 3-17
days 3-24
days 3-30


d
Size works both ways (day 30, n=155)
% Not Acceptable
61%
0%
10%
3%
23%
smallest
largest
Day-30 size fifth


e
Gain over two-way size + plate
CNN std
CNN strong
CNN bbox
CNN br. bbox
LSTM last
LSTM mean
Δ AUC
Day (LSTM: last day of window)


f
With size held fixed, CNNs are near chance
CNN standard
CNN strong aug
CNN bbox crop
CNN bright. bbox
AUC within same-size pairs (±10%)
Day


g
Bright. bbox CNN, day 30: small or big = bad
Acceptable
Not Acceptable
CNN P(Acceptable), OOF mean
Day-30 organoid area (×1000 px)